



The University of Azad Jammu & Kashmir, Muzaffarabad

Computer Architecture & Logic Design

Lab08: SR Latch

Student Name:

Shahzad Ahmed Awan

Roll No:

2024-SE-15

Course Title:

CL&D

Course Code:

CS-1206

Instructor:

Engr. Sidra Rafique

Submission Date:

August-2025

Department of Software Engineering

1. Objective

The aim of this lab is to design, simulate, and analyze the behavior of an SR Latch using basic logic gates. The experiment helps to understand its operation under various input combinations and verify its functionality using simulation tools.

The aim of this lab is to design, simulate, and analyze the behavior of an **SR Latch** using basic logic gates. The experiment helps to understand its operation under various input combinations and verify its functionality using simulation tools.

2. S-R Latches

The SR (Set-Reset) Latch is a fundamental sequential circuit that can store a single bit of binary information. It is the simplest memory element and forms the foundation for more complex flip-flops and storage devices.

► Key Characteristics:

- **Inputs:**
 - **S (Set):** Sets output Q to 1
 - **R (Reset):** Resets output Q to 0
- **Outputs:**
 - **Q:** Main output
 - **$\bar{Q}$ (Q bar):** Complement of Q

► Logic Implementation:

- Can be built using either NOR gates or NAND gates.
- In this lab, the SR latch is implemented using NOR gates.

► Working Principle:

- The circuit uses feedback from the outputs to the inputs to hold its state.
- When both inputs are LOW ($S = 0, R = 0$), the latch remembers the last state.
- When $S = 1$ and $R = 0$, output Q is set to 1.
- When $S = 0$ and $R = 1$, output Q is reset to 0.
- When $S = 1$ and $R = 1$ (both HIGH), it leads to an invalid or undefined condition in NOR-based latches.

4. Implementation Using NOR Gates

A. Truth Table (NOR-based SR Latch)

S	R	Q (Next State)	$\bar{Q}$	Operation
0	0	No Change	No Change	Memory / Hold
0	1	0	1	Reset
1	0	1	0	Set
1	1	Invalid	Invalid	Forbidden / Invalid

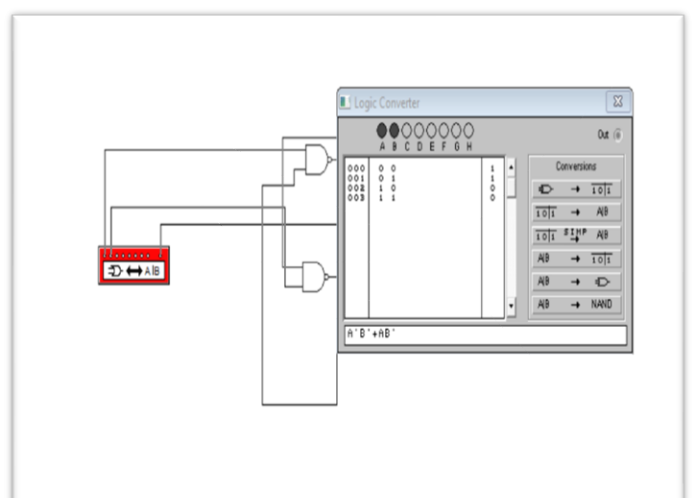
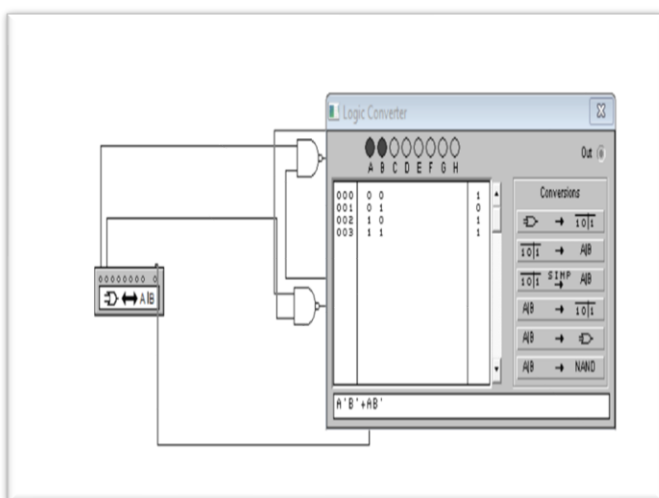
5. Implementation Using NAND Gates

A. Truth Table (NAND-based SR Latch)

S	R	Q (Next State)	$\bar{Q}$	Operation
1	1	No Change	No Change	Memory / Hold
1	0	1	0	Set
0	1	0	1	Reset
0	0	Invalid	Invalid	Forbidden / Invalid

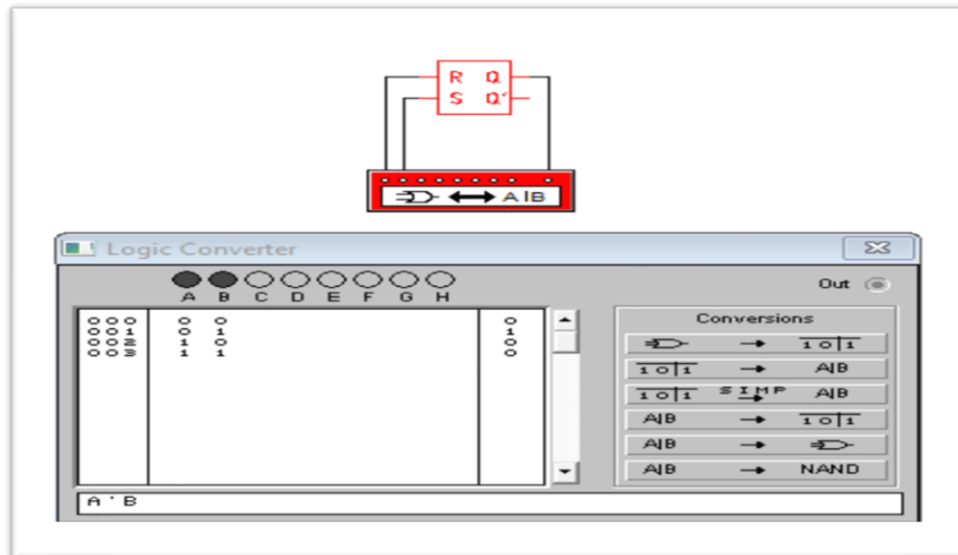
In NAND-based latches, logic is **active-low**, so **1** is **idle**, and control is done by pulling inputs to **0**.

Implementation in EWB



Using Built-in:

Truth Table of Q



Truth Table of Q'

